

Course Outline

Software Integration and Maintenance

Prepared from the official course profile

Item	Details
Course Code	0613-412
Course Title	Software Integration and Maintenance
Course Type	Core Course
Year/Semester	7th Semester
Credit Value	3.0
Contact Hours	3 hours per week
Total Marks	100
Proposed Teaching Method	Theoretical lectures with Q/A, short demonstration, group discussion, and revision sessions

1. Course Rationale

This course develops students' understanding of software integration, software maintenance, testing strategies, testing tools, and performance measurement. It emphasizes how integrated software systems are planned, maintained, tested, measured, and improved throughout the system lifecycle.

2. Course Objectives

- To familiarize students with advanced integration technologies used in software integration solutions.
- To help students understand detailed requirements, metrics, and the impact of requirement changes.
- To introduce practical integration issues through examples and case-based discussion.
- To develop students' ability to measure testing progress and evaluate testing outcomes.
- To support the preparation of effective test plans and maintenance strategies across the software lifecycle.

3. Course Learning Outcomes

CLO	Expected Learning Outcome	Bloom Level	Mapped Assessment
CLO1	Describe advanced integration technologies to implement software integration solutions.	Understand	Class Test, Presentation, Mid-Term
CLO2	Analyze tests for limited security vulnerabilities and automated testing tools.	Analyze	Q/A, Demonstration, Assignment
CLO3	Develop effective test plans and maintenance strategies throughout the system lifecycle.	Create	Assignment, Final Examination

4. Proposed Course Outline

Week	Lecture Topic	Teaching Method	Assessment/Activity
1	Course introduction; overview of software integration and maintenance; course outcomes and assessment plan.	Lecture, Q/A	Introductory Q/A
2	Software integration: types of data needing integration, APIs, middleware, integration platforms, benefits and challenges.	Lecture, examples	Short class discussion
3	Software maintenance: need, process, cost, advantages, disadvantages, and factors affecting maintenance cost.	Lecture	Class Test preparation
4	Maintenance types: corrective, adaptive, perfective, preventive; maintenance activities; reengineering and legacy systems.	Lecture, case discussion	Assignment briefing
5	SCM and change management: version control, semantic versioning, configuration auditing, status reporting.	Lecture, demonstration	Q/A
6	Integration strategy: big-bang, top-down, bottom-up, hybrid/sandwich integration; drivers and stubs.	Lecture	Class Test (CT-1)
7	Software Testing Life Cycle, testing principles, testing methodology, test approaches, and automation benefits.	Lecture	Mid-Term preparation
8	Mid-Term Examination.	Written examination	Mid-Term
9	Types of testing: alpha, beta, agile, ad hoc, black-box, white-box, stress, and application testing.	Lecture, examples	Q/A
10	Integration testing: purpose, advantages, challenges, top-down, bottom-up, entry criteria, exit criteria, and integration test cases.	Lecture, template discussion	Assignment progress check
11	Testing tools: manual, visual, automation, functional, GUI, penetration, performance, SOA, accessibility, QTP/UFT, regression, security, and test management tools.	Lecture, tool overview	Class Test (CT-2)

Week	Lecture Topic	Teaching Method	Assessment/Activity
12	Security lifecycle, reliability, operational profile, performance/load/stress testing.	Lecture	Q/A
13	Test progress metrics, measurements and reports, test effort estimation, risk assessment and mitigation.	Lecture, problem examples	Assignment submission
14	Revision, course review, question solving, and final examination preparation.	Revision, group discussion	Final exam preparation

5. Assessment Plan

Assessment Component	Suggested Weight	Purpose
Attendance + Behaviour	10%	To evaluate students' regular class attendance, punctuality, classroom discipline, participation, and professional academic behaviour throughout the semester.
Class Test (CT)	10%	Evaluate continuous understanding of integration, maintenance, testing strategies, and testing tools.
Mid-Term Examination	20%	Assess first-half topics: software integration, software maintenance, SCM, and integration strategy.
Assignment	10%	Evaluate students' ability to prepare a short academic/practical report on integration, maintenance strategy, testing tools, or testing documentation.
Final Examination	50%	Comprehensively assess the full syllabus, including testing, integration testing, tools, metrics, and maintenance measurement.

6. Assignment Topics

- Prepare a software integration plan for a university management system.
- Compare top-down, bottom-up, big-bang, and hybrid integration strategies using examples.
- Design an integration test plan with entry criteria, exit criteria, test cases, and defect report format.
- Compare selected software testing tools such as Selenium, Playwright, JMeter, Burp Suite, Postman, and TestRail.
- Prepare a maintenance strategy for a legacy software system, including reengineering decisions.

7. References:

1. Penny Grubb, Armstrong A. Takang, "Software Maintenance: Concepts and Practice", World Scientific, 2nd Edition, 2017.
2. Dr. (Prof.) Rajendra Prasad, Prof. Govind Verma, "Software Engineering", Khanna Publishing, 1st Edition, 2016.